

A Minimal and Jitter-Free FPGA Telemetry Core for Cycle-Accurate Determinism in Real-Time Systems

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Abstract—This work presents a minimal hardware-only FPGA telemetry core designed for deterministic real-time monitoring. By replacing interrupt-driven software execution with a finite-state-machine pipeline, the architecture achieves a fixed latency of four clock cycles with zero timing jitter. HDL simulation and waveform analysis show a measured internal latency of 20 ns between data-ready assertion and UART transmission start. Implemented on a Xilinx Artix-7 FPGA, the design occupies only 49 LUTs, making it suitable for compact cyber-physical systems where predictable timing is more critical than throughput.

I. INTRODUCTION

Real-time telemetry systems used in aerospace, robotics, and industrial monitoring require strict temporal determinism. Software-driven communication stacks introduce unpredictable latency due to interrupt handling and task scheduling. FPGA-based hardware pipelines eliminate these sources of variability by executing communication logic directly in hardware.

This work proposes a minimal deterministic telemetry communication core implemented entirely in FPGA logic. The architecture focuses on cycle-accurate latency rather than protocol complexity or throughput.

II. ARCHITECTURE

The telemetry core is implemented as a standalone hardware pipeline inside the FPGA fabric. A finite state machine (FSM) governs the telemetry flow through four deterministic states: *IDLE*, *LOAD*, *FRAME*, and *SEND*. Because all transitions are synchronized with the system clock, the telemetry pipeline operates with strictly predictable timing behavior.

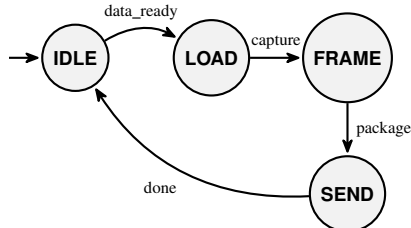


Fig. 1: FSM-based telemetry control architecture.

III. DETERMINISTIC TIMING

Deterministic timing behavior was verified using HDL simulation. The telemetry pipeline exhibits a fixed latency independent of data payload.

- Internal latency: 4 clock cycles
- Measured latency: 20 ns
- Timing jitter: 0

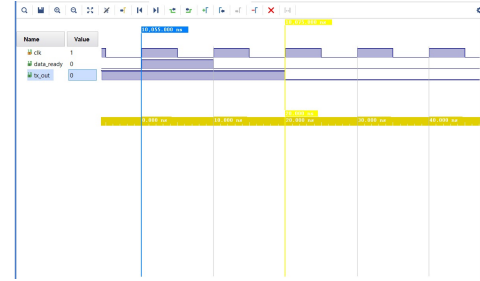


Fig. 2: Waveform verification showing fixed 4-cycle latency (20 ns) between data_ready assertion and UART transmission start.

IV. EXPERIMENTAL VALIDATION

The telemetry core was implemented on a Digilent Nexys A7 (Artix-7 XC7A100T) FPGA board. Logic analyzer measurements confirmed the constant latency between the data acquisition event and the UART transmission trigger. Across multiple simulation runs and experimental tests, the architecture maintained a strictly constant four-cycle latency with zero observable timing jitter.

Compared with typical microcontroller-based telemetry systems where interrupt-driven communication introduces microsecond-level variability, the proposed hardware-only design provides deterministic timing behavior with a minimal hardware footprint of only 49 LUTs.

V. CONCLUSION

This work demonstrated a lightweight FPGA telemetry core capable of deterministic communication with fixed cycle-accurate latency. The architecture eliminates software-induced timing variability by implementing the entire telemetry pipeline in hardware using an FSM-driven design. With only 49 LUTs and a measured latency of 20 ns, the proposed design is suitable for real-time monitoring applications where temporal determinism is critical.

REFERENCES

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